Smart Driver Safety System

Image-processing project

C05

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Problem Statement

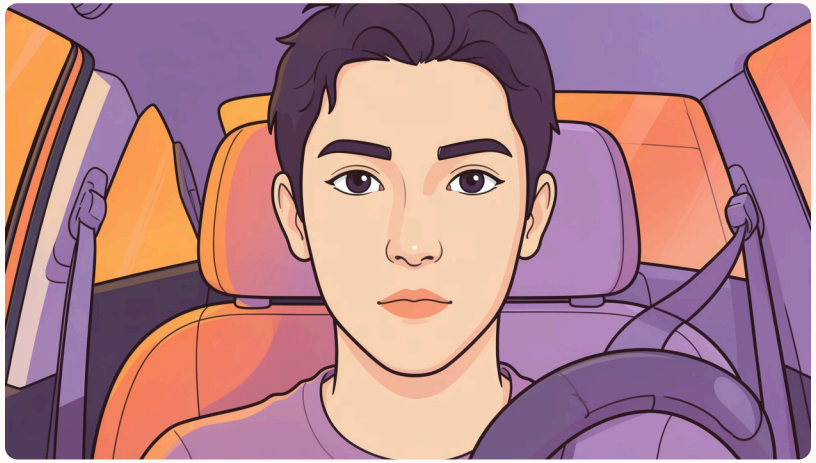
PROBLEM STATEMENT

Driver drowsiness is one of the main causes of road accidents.

A reliable system is needed to automatically detect whether a driver is:

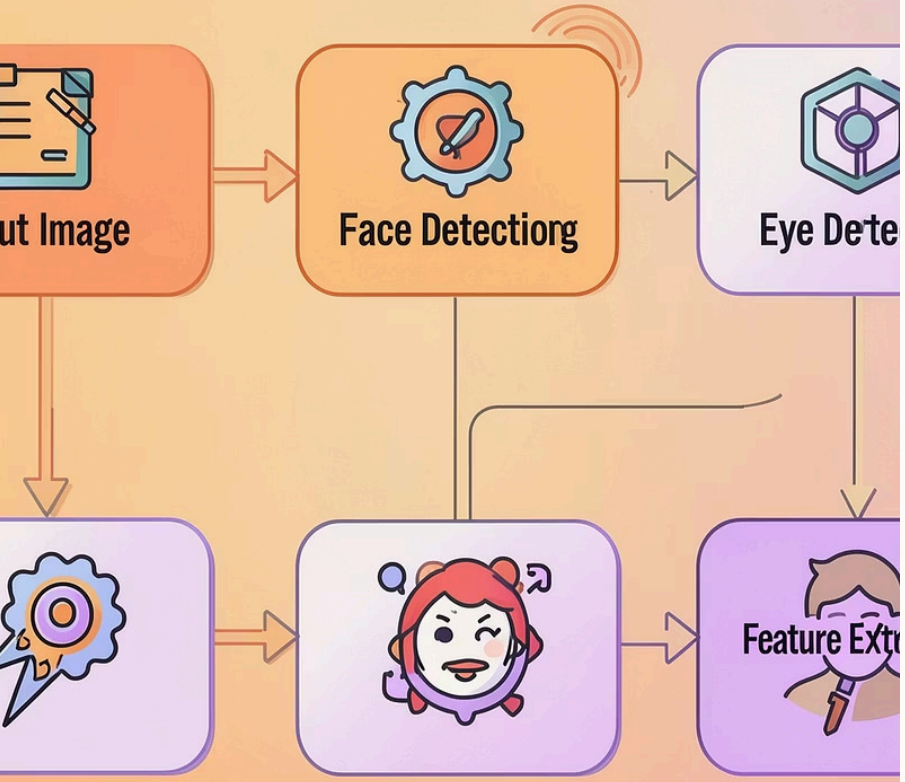
- Drowsy
- Non-Drowsy

⚠ Goal: Reduce crash risk by enabling timely alerts or interventions.



Project Objectives

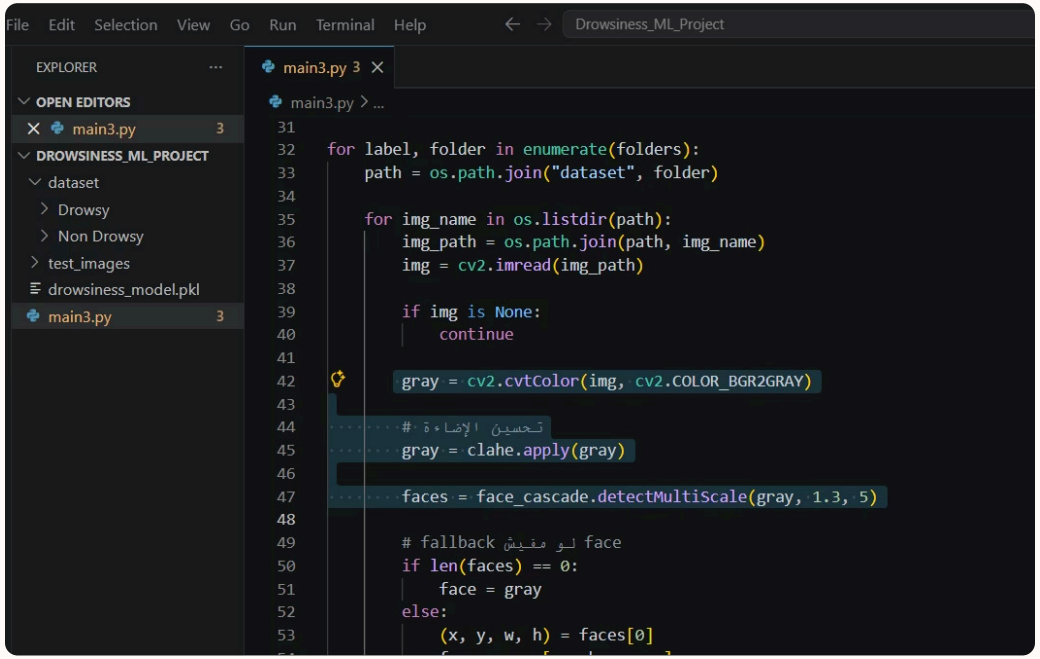
- The main objective of this project is to:
- Detect driver drowsiness using image processing
- Build a complete computer vision system
- Improve road safety



System Pipeline (High-level)

Input Image → Preprocessing → Face Detection → Eye Detection → Feature Extraction → Classification (SVM) → Output (Drowsy / Non-Drowsy)

Each stage is modular to allow component upgrades (e.g., replace SVM with CNN later).

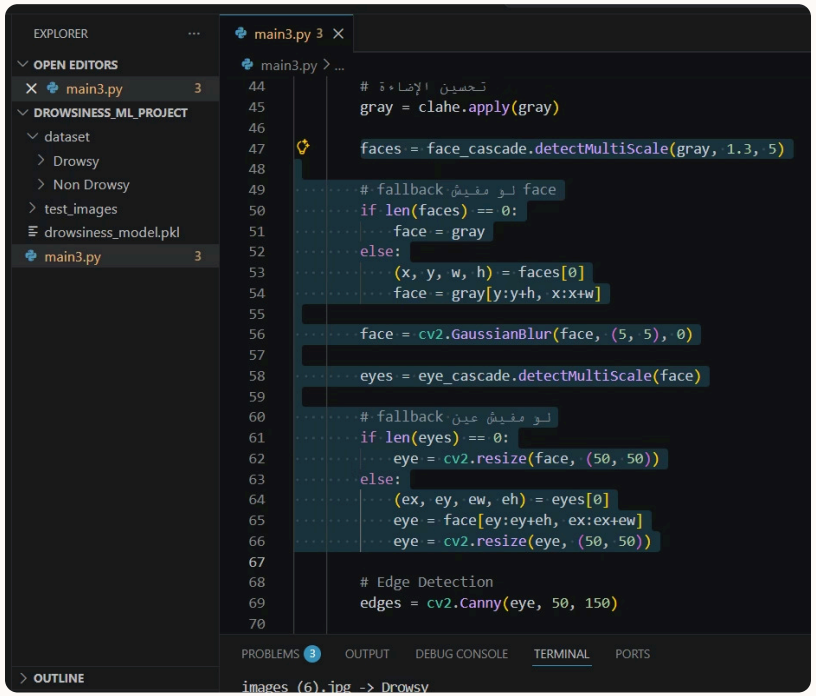


Preprocessing

- Techniques applied:
- Grayscale conversion
- CLAHE (improves lighting conditions)
- Gaussian Blur (reduces noise)

Purpose:

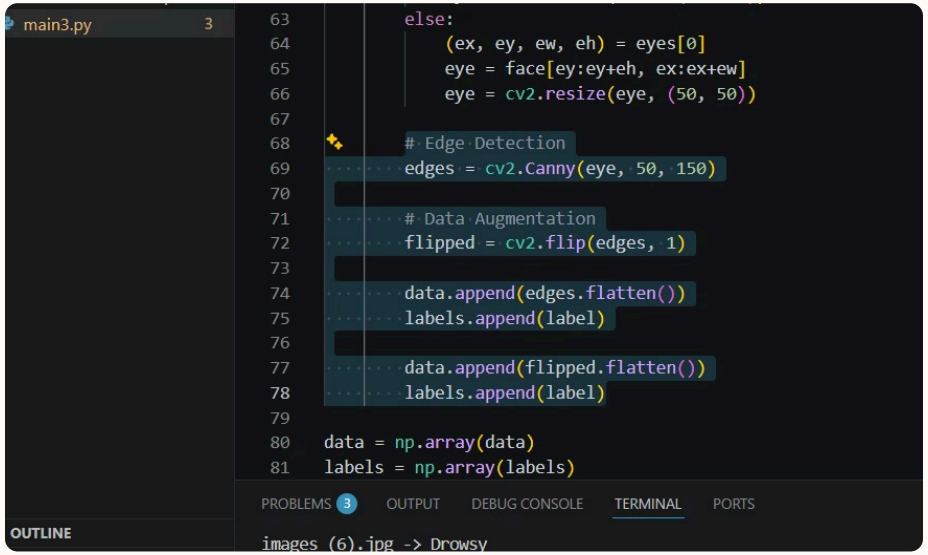
- Enhance image quality
- Improve detection accuracy



Detection Strategy

- **Face detection**: Haar cascade classifier — lightweight and real-time capable
- **Eye detection**: localized search within detected face region to reduce false positives
- **Fallbacks**:
  - If no face → use full image

    If no eyes → use face region



Feature Extraction

- Resize image to 50×50
- Apply Edge Detection (Canny)
- Convert image to feature vector

Purpose:

- Convert image into numerical data feature vectors for fast SVM inference and smaller training sets.

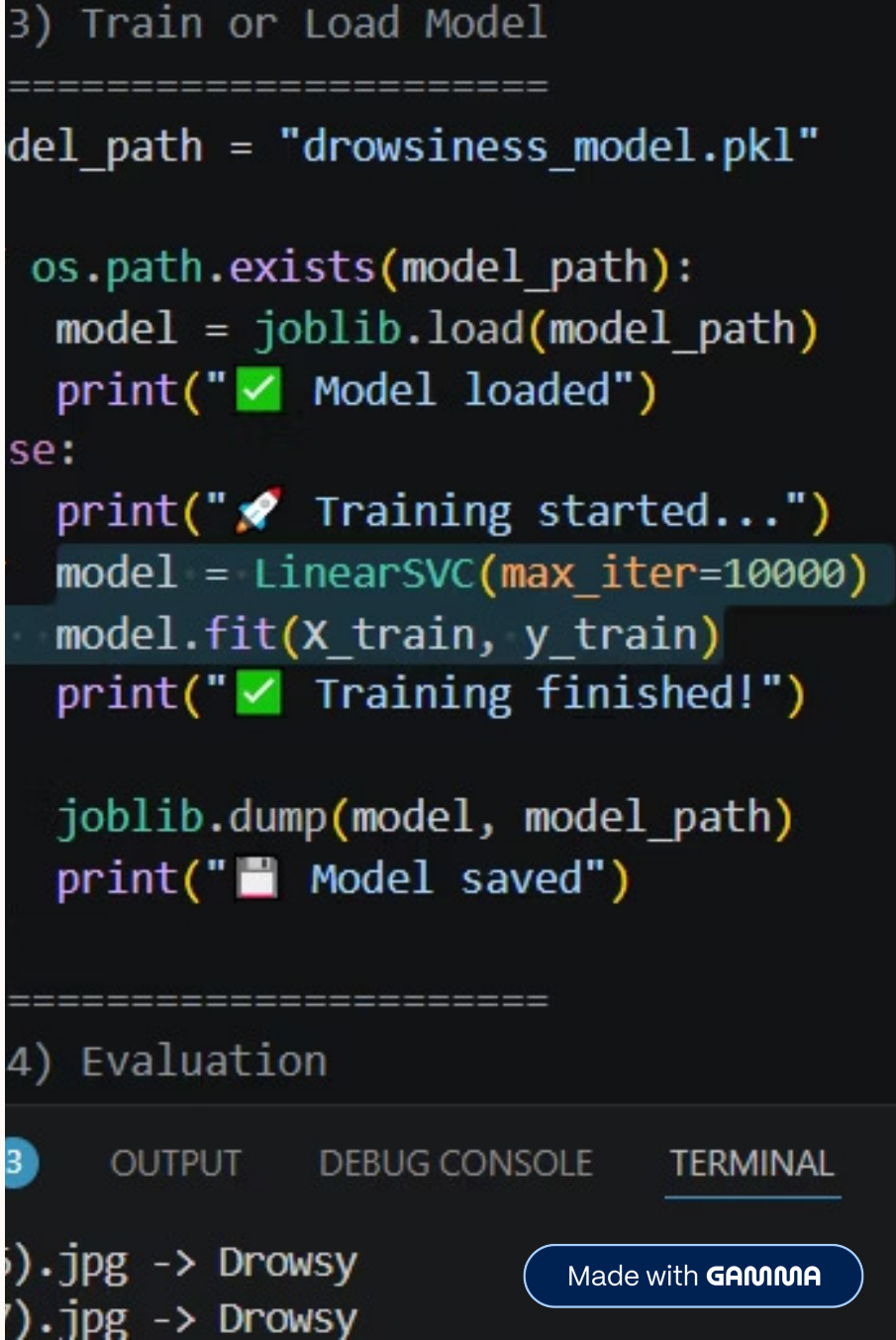
Classification

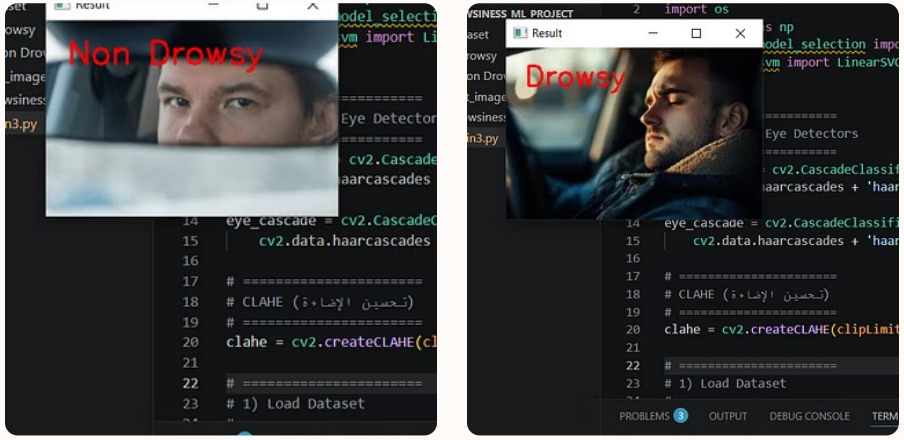
Model Used:

- Support Vector Machine (LinearSVC)

Output:

- Drowsy
- Non-Drowsy.





Results — Detection & Examples

The system:

- Detects face
- Detects eye
- Displays classification result

Examples:

- Drowsy
- Non-Drowsy

(Insert images here).

COMPARISON

A comparison was performed between:

Method	Accuracy
Without CLAHE	~0.46
With CLAHE	~0.50

Conclusion:

- CLAHE improves contrast
- Enhances system performance

CHALLENGES

The system faces some limitations:

- Lighting variations
- Face angle differences
- Eye detection failure in some cases

FUTURE WORK

Possible improvements:

- Use CNN (Deep Learning)
- Increase dataset size
- Real-time detection using camera

CONCLUSION

We successfully developed a complete system that:

- Uses multiple image processing techniques
- Applies machine learning for classification
- Provides real-time-like results